Problem 0

Let

$$\mathbf{v} = \langle 1, \sqrt{3} \rangle, \qquad \mathbf{r}(t) = \left\langle \sin\left(\frac{\pi t}{6}\right), \cos\left(\frac{\pi t}{6}\right) \right\rangle.$$

(a) Solve the system of equations $\lambda \mathbf{r}(t) = \mathbf{v}$.

(b) Interpret your answer to part (a) geometrically.

Problem 1

In a simplified model, a planet at distance r from its star receives radiation at a rate per unit area

$$F(r) = \frac{L}{4\pi r^2}.$$

Its equilibrium temperature is

$$T(F, A) = \left(\frac{(1 - A)F}{4\sigma} \right)^{1/4},$$

where A is the fraction of incoming radiation the planet reflects, L is the star's luminosity, and σ is the Stefan–Boltzmann constant, with L and σ held constant.

(a) Compute $\dfrac{\partial T}{\partial A}$ and $\dfrac{\partial T}{\partial r}$.

(b) With the star at the origin, $r(x, y) = \sqrt{x^2 + y^2}$. For fixed A, find the direction of greatest temperature increase at $(0, 3)$.

(c) With A fixed, find the instantaneous rate of change of temperature at $(0, 3)$ if the planet has velocity $\langle 1, 2 \rangle$.

Problem 2

Let

$$f(x, y, z) = x^2 + y^2 + z^2 - 2x - 4y - 2z.$$

(a) Find the absolute maximum and minimum of f on the region $x \geq 0$, $y \geq 0$, $z \geq 0$, or explain why they do not exist.

(b) Repeat part (a) for the region $x, y, z \leq 0$.

Problem 3

A small workshop's daily output is modeled by

$$Q(x, y) = xy^2,$$

where x is the number of hours of labor and y is the number of hours of machine time. Labor costs \$20 per hour, and machine time costs \$10 per hour. Spending its entire daily budget of \$240 subjects the workshop to the constraint

$$20x + 10y = 240.$$

(a) Find the allocation of labor and machine time that maximizes daily output, and find that output.

(b) If the daily budget doubles, find the new optimal allocation and maximum output. Does the maximum output also double?

Problem 0: Answer Key

(a) In components, the system is

$$\lambda \sin\left(\frac{\pi t}{6}\right) = 1, \qquad \lambda \cos\left(\frac{\pi t}{6}\right) = \sqrt{3}.$$

Squaring and adding gives $\lambda^2 = 1 + 3 = 4$, so $\lambda = \pm 2$.

For $\lambda = 2$,

$$\sin\left(\frac{\pi t}{6}\right) = \frac{1}{2}, \qquad \cos\left(\frac{\pi t}{6}\right) = \frac{\sqrt{3}}{2},$$

hence $\pi t/6 = \pi/6 + 2\pi k$. For $\lambda = -2$, both signs reverse, giving $\pi t/6 = 7\pi/6 + 2\pi k$. All solutions are

$$\boxed{(\lambda, t) = (2,\ 1 + 12k) \quad \text{or} \quad (-2,\ 7 + 12k), \qquad k \in \mathbb{Z}.}$$

(b) The equation says that $\mathbf{v}$ is a scalar multiple of $\mathbf{r}(t)$, so **the two vectors must be parallel**. Since $\|\mathbf{r}(t)\| = 1$ and $\|\mathbf{v}\| = 2$, the scale factor has magnitude 2: $\lambda = 2$ when they point in the same direction and $\lambda = -2$ when they point in opposite directions. Like a clock's hour hand, $\mathbf{r}(t)$ points toward 1 o'clock when $t = 1$ and toward 7 o'clock when $t = 7$, completing one revolution every 12 hours when t is measured in hours.

Problem 1: Answer Key

(a) Substituting $F(r)$ gives

$$T(r, A) = \left(\frac{L}{16\pi\sigma}\right)^{1/4} (1-A)^{1/4} r^{-1/2}.$$

Therefore,

$$\boxed{\frac{\partial T}{\partial A} = -\frac{1}{4}\left(\frac{L}{16\pi\sigma}\right)^{1/4} (1-A)^{-3/4} r^{-1/2}},$$

$$\boxed{\frac{\partial T}{\partial r} = -\frac{1}{2}\left(\frac{(1-A)L}{16\pi\sigma}\right)^{1/4} r^{-3/2}}.$$

(b) For fixed A, the chain rule gives

$$\frac{\partial T}{\partial x} = \frac{\partial T}{\partial r}\frac{\partial r}{\partial x} = -\frac{x}{2(x^2+y^2)^{5/4}}\left(\frac{(1-A)L}{16\pi\sigma}\right)^{1/4},$$
$$\frac{\partial T}{\partial y} = \frac{\partial T}{\partial r}\frac{\partial r}{\partial y} = -\frac{y}{2(x^2+y^2)^{5/4}}\left(\frac{(1-A)L}{16\pi\sigma}\right)^{1/4}.$$

Thus,

$$\nabla T(0,3) = \left\langle 0, -\frac{1}{6\sqrt{3}}\left(\frac{(1-A)L}{16\pi\sigma}\right)^{1/4} \right\rangle.$$

Normalizing the gradient gives the direction of greatest increase:

$$\boxed{\frac{\nabla T(0,3)}{\|\nabla T(0,3)\|} = \langle 0, -1\rangle}.$$

(c) Along the planet's path,

$$\frac{dT}{dt} = \frac{\partial T}{\partial x}\frac{dx}{dt} + \frac{\partial T}{\partial y}\frac{dy}{dt}.$$

At the given instant, this becomes

$$\frac{dT}{dt} = 0(1) - \frac{1}{6\sqrt{3}}\left(\frac{(1-A)L}{16\pi\sigma}\right)^{1/4}(2) = \boxed{-\frac{1}{3\sqrt{3}}\left(\frac{(1-A)L}{16\pi\sigma}\right)^{1/4}}.$$

Problem 2: Answer Key

(a) The critical-point equations are

$$f_x = 2x - 2 = 0, \qquad f_y = 2y - 4 = 0, \qquad \frac{\partial f}{\partial z} = 2z - 2 = 0.$$

They give $(1, 2, 1)$, which lies in the region. Completing squares gives

$$f(x, y, z) = (x - 1)^2 + (y - 2)^2 + (z - 1)^2 - 6 \geq -6.$$

Equality holds only at $(1, 2, 1)$, so the absolute minimum is

$$\boxed{-6 \text{ at } (1, 2, 1)}.$$

Along $(t, 0, 0)$ with $t \geq 0$,

$$f(t, 0, 0) = t^2 - 2t \longrightarrow \infty \quad \text{as } t \longrightarrow \infty.$$

Thus $\boxed{\text{no absolute maximum exists}}$.

(b) When $x, y, z \leq 0$, all six terms in

$$f(x, y, z) = x^2 + y^2 + z^2 - 2x - 4y - 2z$$

are nonnegative. Their sum equals zero only at the origin, so the absolute minimum is

$$\boxed{0 \text{ at } (0, 0, 0)}.$$

Along $(-t, 0, 0)$ with $t \geq 0$,

$$f(-t, 0, 0) = t^2 + 2t \longrightarrow \infty \quad \text{as } t \longrightarrow \infty.$$

Thus $\boxed{\text{no absolute maximum exists}}$.

Problem 3: Answer Key

(a) The budget constraint is $20x + 10y = 240$. At an interior constrained extremum, the Lagrange equations are

$$y^2 = 20\lambda, \qquad 2xy = 10\lambda, \qquad 20x + 10y = 240.$$

Eliminating λ gives $y^2 = 4xy$, or $y(y - 4x) = 0$. At an interior point, $y > 0$, so $y = 4x$. Substitution into the constraint gives $60x = 240$. Thus the optimal allocation and output are

$$\boxed{x = 4, \qquad y = 16, \qquad Q(4, 16) = 4(16)^2 = 1024}.$$

The feasible set is the closed line segment from $(0, 24)$ to $(12, 0)$. Since Q is continuous, it attains an absolute maximum on this segment. The only interior candidate is $(4, 16)$, where $Q = 1024$, and both endpoints give $Q = 0$. The case $y = 0$ from part (a) is the endpoint $(12, 0)$. Therefore $(4, 16)$ gives the absolute maximum.

(b) With a budget of \$480, the constraint becomes $20x + 10y = 480$. The interior Lagrange equations still give $y = 4x$, so $60x = 480$ and

$$\boxed{x = 8, \qquad y = 32, \qquad Q(8, 32) = 8(32)^2 = 8192}.$$

Both endpoints again give zero output, so this is the absolute maximum. The maximum output increases by a factor of **eight**, since

$$Q(2x, 2y) = (2x)(2y)^2 = 8xy^2 = 8Q(x, y).$$